

Use-Case Flow Specification

Use-Case ID	UC-02
Use-Case Name	Dispense Medicine (FIFO Checkout)
Primary Actor	Pharmacy Clerk
Trigger	The Pharmacy Clerk submits a prescription or counter sale for checkout.
Related Requirements	FR-001 (FIFO ordering), FR-004 (threshold re-evaluation), NFR-002 (audit trail)

Preconditions

1. The Pharmacy Clerk is authenticated and holds the DISPENSE permission.
2. A valid prescription or counter-sale record exists, listing medicine codes and quantities.
3. At least one batch of each requested medicine exists with status AVAILABLE.
4. The daily expiry sweep (UC-03) has completed for the current date, so batch statuses are current.

Postconditions

1. The requested quantity is deducted from the nearest-expiry AVAILABLE batch(es) of each medicine.
2. The stock ledger is updated and an immutable audit entry is written for every deduction, attributed to the clerk and timestamped (NFR-002).
3. Dispensable stock for each affected item is re-evaluated against its safety threshold; a re-order trigger is raised where the threshold has been breached (FR-004).
4. A dispense slip is generated listing, per line, the medicine, quantity, batch ID and expiry date issued.

On failure: No stock is deducted, no ledger or audit entry is written, and the sale remains open — a dispense transaction is all-or-nothing across every line in the sale.

Main Success Scenario

Step	Actor	Action
1	Clerk	Opens the checkout screen and selects the prescription or sale record to dispense.
2	System	Displays each requested medicine with the quantity to be dispensed.
3	Clerk	Confirms the dispense request.
4	System	For each line, retrieves all batches of that medicine with status AVAILABLE and sorts them by expiry date ascending, breaking ties by earliest registration date.
5	System	«include» UC-07 Validate Batch Expiry — re-checks the selected batch against the current system date to confirm it has not expired since the last sweep.
6	System	Allocates the requested quantity to the nearest-expiry valid batch and displays the proposed allocation (medicine, batch ID, expiry date, quantity) for confirmation.
7	Clerk	Confirms the allocation.
8	System	«include» UC-08 Update Stock Ledger — deducts the allocated quantity, writes the ledger movement and the audit entry, and commits the transaction.
9	System	«include» UC-10 Evaluate Stock Threshold — compares remaining dispensable stock against the configured safety threshold and raises a re-order trigger if it has been breached.
10	System	Generates and prints the dispense slip, and returns the clerk to the checkout screen.

Alternate Flow — 6a. Nearest-expiry batch holds insufficient quantity

Step	Actor	Action
6a.1	System	Detects that the nearest-expiry valid batch holds less than the requested quantity.
6a.2	System	Allocates the full remaining quantity of that batch, then moves to the next batch in ascending expiry order and repeats until the requested quantity is fully covered, preserving FIFO order across the split.
6a.3	System	Displays the multi-batch allocation, showing the quantity drawn from each batch with its batch ID and expiry date.
6a.4	Clerk	Confirms the split allocation. The flow rejoins the main success scenario at step 8, where all deductions in the split are committed as a single atomic transaction.
6a.5	System	If the combined stock across all valid batches is still short, dispenses nothing, informs the clerk of the maximum dispensable quantity, and raises a re-order trigger for the item (FR-004).